Iteration and inquiry: Toward a meaningful model of ethical engagement for engineering and computing students

Alexandra Morrison
Humanities Department
Michigan Technological University
Houghton, USA
lamorris@mtu.edu

Charles Wallace
Computer Science Department
Michigan Technological University
Houghton, USA
wallace@mtu.edu

2021 IEEE International Symposium on Ethics in Engineering, Science and Technology (ETHICS) | 978-1-6654-0801-1/21/$31.00 ©2021 IEEE | DOI: 10.1109/ETHICS53270.2021.9632741

***Abstract*—Establishing an appropriate foundation for teaching ethics in engineering and computing disciplines can be a profound challenge. The essence of these disciplines lies in building and doing, but how does one "build" or "do" ethics? Without a basis for putting ethical reasoning into practice, even well-intentioned curricular efforts can perpetuate a sense that ethics is somehow marginal to students' core discipline. We have used the Ethical Cycle, developed by philosophers van de Poel and Royakkers, as a way to address this problem. The Ethical Cycle is an operational model with well-defined stages: statement of the moral problem, analysis, identification of options for action, ethical evaluation, and reflection. We discuss similarities between components of the Ethical Cycle and corresponding practices in agile software development. These similarities can help students feel more confident and motivated to reason critically about ethical concerns. We also point out some risks of conflating the two: a reductionist perspective that equates ethics with functionality, an instrumentalist perspective that detaches responsibilities as software developers from larger ethical imperatives, and a misapprehension that ethical reasoning inevitably leads to a "solution". We discuss our experiences adopting and adapting the Ethical Cycle for an engineering ethics course, along with supportive material from philosophy of technology and Science and Technology Studies.**

Keywords—ethics education, engineering education, computer science education, Ethical Cycle, philosophy of technology, Science and Technology Studies, agile methods, critical inquiry

I. Introduction

The spate of recent accidents, breaches, and scandals involving high tech—like Cambridge Analytica's data harvesting and psychological profiling, Google's biased algorithms and antitrust violations, Facebook's countless abuses of data and election interference—have brought engineering ethics to the forefront of public debate. While there has always been a public expectation or hope for high tech companies and other institutions—and by extension their employees—to act ethically, the profound and immediate effects of recent corporate ethical lapses on the public have sparked efforts to build ethics into the workplace in a more explicit way than before, as can been seen with the high-profile examples of ethics advisory boards springing up across the country. Similarly, there has been heightened interest in preparing engineering students to be ethical decision makers [1][2]. Ironically, however, with these apparent markers of progress comes a new uncertainty: what do the seemingly self-evident concepts of ethics and ethical practice mean in a high-tech context? This new uncertainty is perhaps unsurprising; as Moss and Metcalf note, "now that 'ethics' is a site of power, who gets to determine the meaning and practices of 'ethics' is being hotly contested [3]."

At a time when what it means to be ethical in the professional sphere is under scrutiny, we should be asking similar questions about ethics education. Certain institutional realities make it profoundly challenging to establish an appropriate and engaging foundation for teaching ethics in computing and engineering. The values of "technological solutionism" and "market fundamentalism" [4] that are ingrained in high tech culture have likewise taken root in educational instructions, curricula and instruction [5]. Moreover, engineering educators tend to have little background in ethical philosophy and hence limited experience with true ethical inquiry [6]; without time or resources to step out of their disciplines and learn, it is not surprising that they tend to replicate the uncritical approaches to ethics education that they encountered as students.

Consequently, like the situation in the tech industry, visions of ethical competencies among engineering students all too often have a reactive quality, as in the recent reformulation in the ABET accreditation requirements for U.S. institutions, foregrounding ethical reasoning as a "Student Outcome". More generally, ABET calls for "an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts". Riley notes that such definitions, framed within a context of "compliance" and intended to delineate a "bare minimum" of competency, should not be confused with goals for excellent ethics education [7]. Furthermore, compliance-based definitions can all too easily be misinterpreted as dictating a compliance approach to ethics itself. What does a meaningful, teachable practice for students look like? How do we make space for students to engage critically and self-reflectively in the manner necessary for ethical engagement, in the context of technical classes that seem to actively work against such engagement?

Our interest in the Ethical Cycle approach, developed by the philosophers Ibo van de Poel and Lamber Royakkers [8], is motivated by our recognition that the sort of critical inquiry necessary for cultivating ethical responsibility is a kind of

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functional literacy, a set of culturally situated habits of critical inquiry through which students engage with the world. The Ethical Cycle is an iterative, collaborative process that encourages students to reevaluate, with the aid of ethical theory and logical argumentation, their initial "common sense" assumptions about the ethical scenarios they are considering.

II. Learning Context and Goals

At our home institution, like many others, ethics instruction for STEM students includes episodic exposure within its engineering and computing curricula. Engineering students get a brief exposure in their common first-year engineering survey course, while computer science and software engineering students take a third-year course devoted to ethical and social aspects of computing. While teaching ethics within disciplinary courses can motivate the topic and place it within a meaningful context, it presents a problem: engineering and computing instructors typically have no formal education in ethical philosophy, and the reward systems do not motivate many to seek such training [5]. Hess and Fore, in their survey of ethical interventions in engineering across the US, note that such instruction tends to focus on professional codes and case studies, suggesting that these instructors are lacking a basis for doing anything more creative or meaningful [9].

Students at our institution have an additional option of enrolling in an Engineering Ethics class taught by a philosophy professor—a class that employs many of the more rigorous (theoretically grounded and practical) methods that we will describe here. It is, however, a single elective course offered in the Humanities Department that reaches a relatively small number of students. Within this course, however, the Ethical Cycle approach has shown promise as a relatable way of systematically working through ethical scenarios within engineering and computing courses. Additionally, if one were to implement Ethical Cycle in an interdisciplinary team-teaching setting, it could afford an even more integrated and meaningful approach to ethics education throughout engineering curricula.

Before we discuss the methodological approach of the Ethical Cycle however, we want to briefly address the general content and aims of ethics instruction. One of the main learning goals in STEM ethics is for students to recognize, and behave with the awareness, of the fundamental socio-technical character of reality. That ethics (and other social elements) are inextricable from technical features must be appreciated to subvert the false notion that ethics is peripheral to technique. Ethics is "always already" embedded in everyday habits and practices, not just explicit moral dilemmas and dramatic moments of decision, as suggested by "classic" ethics education cases like the Space Shuttle Challenger and Ford Pinto catastrophes. Since values are embodied in our technological relations and habits, engineering and computing ethics is not simply about "proper" use of technology, but rather more critically about design. Software design frames our perception, actions, and consequently our values in subtle and profound ways. For example, digital platforms that encourage users to make frequent quick and unconsidered judgments about other users habituates unreflective practices. Another example is the increasing use of algorithmic decision making tools that frame human individuals, as wholly captured by statistical models.

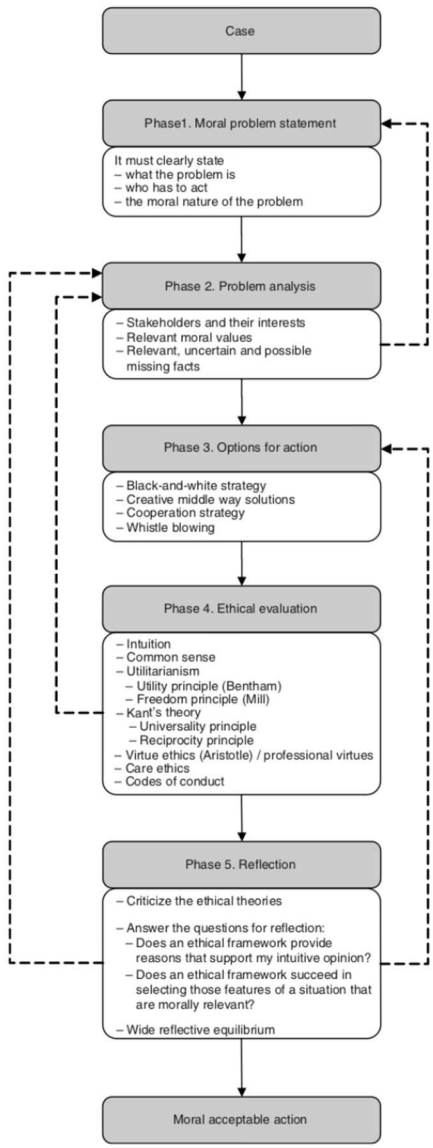

Fig. 1. Ethical Cycle outline [8]

Cultivating critical ethical reasoning skills requires curiosity beyond merely doing the "right" thing at a microethical level, but rather requires engagement with the situation at a macroethical level, necessitating immersion in the broader context. Revealing this context includes arguably much more engagement and collaboration than is typically possible, but even awareness of our limitations on this score can be morally salutary. Battersby and Bailin's description of the aspects of context are helpful reminders of the kinds of considerations necessary for critical ethical inquiry: "(1) knowledge of the dialectical context (the debate around an issue, both current and historical); (2) an understanding of the current state of practice and belief surrounding an issue; (3) an understanding of the intellectual, political, historical and social contexts in which an issue is embedded; (4) knowledge of the relevant disciplinary

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context(s); (5) information about the sources of an argument; (6) awareness of one's own beliefs and biases [10]."

A deeper appreciation of the situated, intersubjective character of agency and moral perception has led to fuller descriptions of ethical experience in contemporary philosophy. This has given rise to more interdisciplinary approaches that put less emphasis on the application of formal rules and more on ways to help students develop sophisticated tools for interpreting complex situations of ethical challenge and moral ambiguity. For example, in engineering ethics it is becoming increasingly common to incorporate other humanities disciplines like philosophy of technology and Science and Technology Studies (STS) [11][12]. These new approaches underscore the myriad ways in which technological interventions, innovations, and designs transform our moral agency. Pedagogy that incorporates these insights more readily enables students to appreciate the complexity of ethical engagement [13].

III. Ethical Cycle

The Ethical Cycle was conceived as a way to adequately address the non-linear or "ill-structured" character of ethical judgement [8][14]. The model could also be used for the training of faculty, instructors and potentially practitioners. Using a series of iterative steps, the Ethical Cycle encourages students to make more deeply considered judgments about real world scenarios. As illustrated in Fig. 1, the five basic steps of the cycle direct students to proceed with moral problem analysis of real-life situations in a structured, iterative manner.

For example, while the cycle begins with a formulation of the moral problem, this problem usually only becomes clear after inquiry into the many facts of the situation, after finding out who are the relevant stakeholders, and after consulting different ethical theories. Formulating a rich description of the moral problem, then, is an iterative process that will be carried over into the other steps, with students continually refining their articulation and expression of the scenario. The problem analysis step requires the student to collect as many relevant elements of the moral situation: the interests of the obvious and less obvious stakeholders, the moral values that are relevant to the situation from the perspectives of the different stakeholders, and the relevant facts. Students often find that after going through this stage they need to reformulate their characterization of the moral problem statement. During the options for action step in the cycle, the student considers possible actions that might mitigate the problem. If they are considering a complex real-world scenario, they will need to creatively generate their own options for action. At this point it is often helpful for students to engage in conversations with the stakeholders involved, if they are thinking through an actual real-world problem, or with classmates whose various backgrounds, experiences, and judgements can be brought into consideration.

After selecting the "best" of the possible actions, students then engage in a more formal type of ethical evaluation, putting aside for the time being, as best as they can, their intuitive reactions in order to consider philosophical frameworks in relation to each of the possible options for action. Traditional ethical theories like utilitarianism, deontological ethics, virtue ethics and ethics of care are employed, not as sets of rules for generating a "right" answer, but rather as ways to illuminate issues or reveal oversights. Students may also turn to the insights from philosophy of technology and STS, and to think about additional relevant ethical considerations with regard to the human-technology relations at play in the scenario.

Finally, students are asked to turn to each other for collaborative reflection. The critical aim at this stage is to reflect upon, and ideally produce some coherence among, three types of moral beliefs: 1) considered moral judgments; 2) moral principles, and; 3) background theories, in a "wide reflective equilibrium" [14]. According to the authors, "The background theories include ethical theories, but also other relevant theories… [since these] block the possibility of simply choosing those principles that fit our considered judgments… As we see it, this process is not so much about achieving equilibrium as such, but about arguing for and against different frameworks and so achieving a conclusion that might not be covered by one of the frameworks in isolation [14]."

Encouraging students to cycle through the scenario again with others is critical because others often disagree, and students need to have the experience of potentially having other perspectives enrich their own ethical reflection. This process can also give a decision more legitimacy because it respects the moral autonomy of others by giving them due care. As Van de Poel and Royakkers point out, this is also more "likely to make you more effective [as an engineer] because you will… need the cooperation of others to live by your moral judgments" [14]. In much of traditional ethics, too much emphasis is placed on individual decision, and students forget that their own moral decisions are actions in a world populated by other things, forces, and persons with which those actions must reckon.

Our experience with the Ethical Cycle in the classroom for the past three years has borne out several of Van de Poel and Royakkers' claims about their method. We have found that when students "walk through" the Ethical Cycle together they come to appreciate, in a deeper, more enduring way, the absolute centrality of dialogue and critical inquiry for ethical engagement. Introducing them to ethical theories in the context of these collaborative sessions has also helped to reduce their tendency to treat these theories—like utilitarianism, deontology or virtue ethics—as if they were "camps" or political parties to which they must swear allegiance. Instead, they discover that the various ethical theories (as well as insights from the philosophy of technology and STS) are more valuably understood to function as disruptors and enhancers of their critical perception. We teach ethical theories to demonstrate how different foundational frameworks can significantly alter how one interprets the same situation as well as possible responses. We have been told by students at the end of sessions that if they had not employed the framework of deontology, for example, they would have simply acquiesced to a utilitarian framing of the situation. Many students also note that without the concept of technological mediation or "scripts" that describe how technologies actively shape our perception and thus our interpretation of the world [17] they would have kept assuming that only human beings are capable of embodying values. This is an untenable position because it so flagrantly contradicts their own experience. They have all seen how, over time, habitual engagement with digital platforms and social media subtly but profoundly changes what *matters* to them and what they engage

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with every day—and thus changes *who* they are. In the end, this collaborative work also helps students to see the mundane is part of ethics; as one student remarked, they initially assumed that ethical reflection wasn't necessary for them since they would never be in a position of designing something as consequential as an O-ring for the Space Shuttle!

IV. Ethical Cycle and Agile Development

There are established engineering processes for the development of software products, and these are part of most computer science programs and an essential component of any software engineering program. Agile development methods, in particular, are motivated by projects of high complexity with requirements that are emergent or unstable, making upfront predictions difficult. Since these characteristics are often associated with software projects, much agile practice is in this area, but much of the foundational material on agile methods refers to systems development in general. Agile development favors adaptive and iterative techniques over predictive and monolithic techniques, for two key reasons: first, it is wasteful to invest time and effort in planning efforts that may end up irrelevant as the project context changes; second, in the absence of dependable analytical tools, creating and testing prototype solutions provides needed evidence for evaluation.

Agile development demands a spirit of constant inquiry. The famous Extreme Programming admonition to "embrace change" [18] implies continuous interaction with stakeholders to understand ever-changing requirements, priorities and obstacles. A key component of agile development is continual process improvement, facilitated by periodic reflective activities. The sprint retrospective in the Scrum framework, for instance, is a ritual that encourages critique and creative problem solving. Derby and Larson [19] position the retrospective as a critical link in an iterative cycle of process evolution (Fig. 2). Since the criteria by which "improvement" is assessed are emergent, identifying and questioning them is part of the retrospective process. Indeed, the choice of certain criteria over others has important ethical consequences.

The similarities between agile methods and the Ethical Cycle approach are striking. Van de Poel and Royakkers, citing Herbert Simon, note the fundamentally "ill-structured" nature of moral problems: "There is no one right way to frame the moral problem and there is no one correct response to address the problem… 'it is usually not possible to make a definitive list of all possible alternative options for action–making solutions always in some sense provisional [14].'" Students with prior agile experience have made similar provisional design steps, that are continually evaluated and compared to alternatives. While this prior experience may not entirely eliminate an attitude of "solutionism", it can make students more receptive to the idea that interesting problems may never be truly "solved".

Ethical reasoning and software engineering share an iterative quality, particularly when situated in an agile process, that can provide an entry point into ethics for both educators and students [20][21]. Since agile methods are now widely integrated into the software industry, students acknowledge them as authentic, in the sense of being methods that they will encounter in the workforce. This provides us a pathway for strengthening student perceptions of ethical reasoning as both useful and actionable.

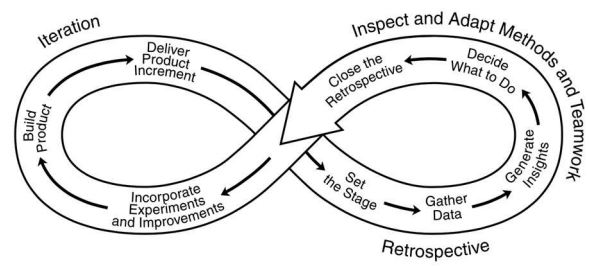

Fig. 2. Agile cycle outline [19]

There are limits, however, in the similarities between the two. First, it is wrongheaded to think of ethical concerns as ordinary software requirements that can be incorporated into a standard development process. Such conflation can lead to a reductionist perspective that equates ethics with functionality, or an instrumentalist perspective that detaches responsibilities as software developers from larger ethical imperatives. Second, iteration in software development is typically an incremental refinement toward a satisfactory implementation that realizes project goals, whereas an iterative ethical process may well call the very foundations of those goals into question. Students may find an iterative model familiar and welcoming, but in ethical reasoning they must be prepared for some iterations of discovery and reflection to take them farther from, not closer to, a "solution". Indeed, failure to produce an implementation at all may be the most ethically sound outcome.

V. Conclusion

Our experiences using the Ethical Cycle in an educational context have demonstrated its promise as an actionable framework that encourages ethical inquiry and critical thinking among engineering students, in a way similar to design practice in the classroom and the workplace. These approaches are not a panacea, but they offer structure to inquiry and critical engagement. They put students in a position of looking from different perspectives, disrupting unexamined beliefs and assumptions. Introducing other theories and frameworks encourages self-reflection and underscores the fact that one's own moral intuitions may not be fully developed, or may be biased) and that critical reflection takes time and collaboration.

An ethics course can provide exposure to the kind of critical inquiry facilitated by the Ethical Cycle, but incorporating it elsewhere in students' engineering experiences would provide the continuous reinforcement needed to make it part of their engineering identity. Introducing opportunities for ethical inquiry into technical project-based courses is promising but challenging; students must be able to balance both technical and ethical cycles of development and inquiry, and engineering educators who facilitate these experiences must themselves feel comfortable with the disruptive nature of iterative ethical reasoning. We conclude with two outstanding questions: how (if at all) can iterative, critical ethical reasoning coexist with established system design processes in engineering courses, and how can engineering programs engage students continuously, rather than episodically, in iterative ethical reasoning?

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